

# Joshua T. Bartkoske

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## EDUCATION

**University of Utah**, Salt Lake City, USA  
*PhD* , Physics - Astronomy track  
Expected Graduation April 2026  
Thesis: "Asteroid occultations with imaging atmospheric Cherenkov telescopes: predictions, observations, and measuring stellar angular diameters."  
Advisor: Dave Kieda

**Montana State University**, Bozeman, USA  
*Bachelor of Science*, Professional Physics  
Graduated May 2020

**Shanghai Ocean University**, Shanghai, China  
*College of International Cultural Exchange*, Chinese Language & Culture Program  
September 2018 - June 2019

**Tien Shan International School**, Almaty, Kazakhstan  
Graduated June 2015

## RESEARCH EXPERIENCE

**University of Utah: Dr. Dave Kieda**  
May 2021 - present

- Analysis of very high energy gamma-ray data: gamma-binary LS I +61 303 and neutrino candidate TXS 0506+056
- Primary researcher on the VERITAS asteroid occultation program: monthly predictions June 2021-present (300+ predictions), analysis of lightcurves (20 detections).
- Observing Experience: 61 nights total at VERITAS in southern Arizona. **7 nights spent as observing czar.** 10 nights spent observing with the stellar intensity interferometry systems at VERITAS.
- Hardware experience: **design** of a prototype Schwarzschild-Couder telescope (pSCT) **camera fixture.** 6 days of camera removal and transport.
- Data quality monitoring: 23 nights for the VERITAS collaboration.

**Montana State University: Dr. Brian D'Urso's Lab**  
Aug 2019 - April 2020

- Loading of microparticles into a magneto-gravitational trap.
- Hardware experience: **built a computer for real-time imaging of microparticles**

**Montana State University: Dr. David Nidever's Research Group**  
Nov 2017 - July 2018

- Analysis of x-ray and radio point sources: cross-matching catalogs of sources in the Large and Small Magellanic Clouds.

**Kazakh British Technical University Research Nanotechnology Laboratory**  
June 2017 - Aug 2017

- Analysis of solar cell anti-reflective coatings: photo-lithography and x-ray reflectivity measurements.

**Montana State University: Dr. Sachiko Tsuruta's Research Group**  
**Oct 2015 - Oct 2017**

- First Year Research Experience: attended group meetings and read scientific papers on neutron stars.

**SERVICE**

- **Member of Organizing Committee for VERITAS Collaboration Meeting** Summer 2025
- AAS Chambliss Poster Evaluator - June 2023
- University of Utah Graduate Assembly Task Force Committee Member Nov 2022 - March 2023
- University of Utah Undergraduate Research Symposium Poster Evaluator - 2022, 2023, 2025
- **Graduate Student Advisory Committee (GSAC) Chair** May 2022 - May 2023
- **GSAC Chair-elect** May 2021 - May 2022
- **Graduate Student Peer Mentor Program Co-Coordinator** 2021, 2022, 2023
- Graduate Student Peer Mentor 2021, 2022, 2023, 2024, 2025
- GSAC Member 2021-2025

**MENTORING**

- Julianna Bayless (2024 Summer REU, University of Missouri) on asteroid occultation secondary analysis.
- Jennifer Yuan (2023 Summer REU, NYU Abu Dhabi) on asteroid occultation angular diameter fits with Gaia DR3 spectra.

**HONORS AND AWARDS**

- **Outstanding Student Presenter 2021** Annual Meeting APS 4 Corners Section
- MSU Award for Excellence Recipient 2020
- Mortar Board 2019
- AP Thompson Scholarship 2019
- Phi Kappa Phi 2019
- Physics Scholarship Recipient 2017
- FYRE Sophomore Scholarship 2016
- The National Society of Leadership and Success 2016
- Mac Rugheimer Memorial Physics Scholarship 2016
- Montana State University Presidential Scholar 2015

**PRIMARY PUBLICATIONS**

- “Direct measurements of the angular diameters of three F-type stars with VERITAS”, VERITAS collaboration, in prep., (target journal: the Astrophysical Journal)
- “VERITAS Asteroid Occultations: Observations and Considerations”, Bartkoske, J. et al., in 39th International Cosmic Ray Conference, 2025, 922, doi: <https://doi.org/10.22323/1.501.0922>

**INVITED TALKS**

- “Resolving Stellar Angular Diameters with Asteroid Occultations”, Seminar, IAA-CSIC, Granada, Spain, Jan 2025

**CONTRIBUTED PUBLICATIONS**

*VERITAS publication author lists are in alphabetical order*

- “Probing a cosmogenic origin of astrophysical neutrinos and cosmic rays using gamma-ray observations of TXS 0506+056”, Acharyya, A. et al., arXiv e-prints, arXiv:2511.06116, 2025. doi:10.48550/arXiv.2511.06116.
- “Constraints on axionlike particles from VERITAS observations of a flaring radio galaxy in the Perseus cluster”, Adams, C. B. et al., Physical Review D, vol. 112, no. 10, Art. no. 103044, APS, 2025. doi:10.1103/jydc-npj6.

- “Multiwavelength Study of HESS J0632+057: New Insights into Pulsar–Disk Interaction”, Park, J. et al., *The Astrophysical Journal*, vol. 991, no. 1, Art. no. 28, IOP, 2025.  
doi:10.3847/1538-4357/adf6b6.
- “Combined dark matter search towards dwarf spheroidal galaxies with Fermi-LAT, HAWC, H.E.S.S. et al., MAGIC, and VERITAS”, LAT Collaboration, arXiv e-prints, arXiv:2508.20229, 2025.  
doi:10.48550/arXiv.2508.20229.
- “Measurement of the photosphere oblateness of  $\gamma$  Cassiopeiae via Stellar Intensity Interferometry with the VERITAS Observatory”, Archer, A. et al., arXiv e-prints, arXiv:2506.15027, 2025.  
doi:10.48550/arXiv.2506.15027.
- “Multiwavelength Observation of a Candidate Pulsar Halo LHAASO J0621+3755 and the First X-Ray Detection of PSR J0622+3749”, Adams, C. B. et al., *The Astrophysical Journal*, vol. 985, no. 1, Art. no. 90, IOP, 2025.  
doi:10.3847/1538-4357/adc92b.
- “Constraints on the X-Ray and Very-high-energy Gamma-Ray Flux from Supernova Remnant W44”, Archer, A. et al., *The Astrophysical Journal*, vol. 983, no. 1, Art. no. 73, IOP, 2025.  
doi:10.3847/1538-4357/adc07d.
- “VERITAS and Multiwavelength Observations of the Blazar B3 2247+381 in Response to an IceCube Neutrino Alert”, Acharyya, A. et al., *The Astrophysical Journal*, vol. 982, no. 2, Art. no. 80, IOP, 2025.  
doi:10.3847/1538-4357/ad30c.
- “An In-depth Study of Gamma Rays from the Starburst Galaxy M82 with VERITAS”, Acharyya, A. et al., *The Astrophysical Journal*, vol. 981, no. 2, Art. no. 189, IOP, 2025.  
doi:10.3847/1538-4357/adab71.
- “Broadband multi-wavelength properties of M87 during the 2018 EHT campaign including a very high energy flaring episode”, Algaba, J. C. et al., *Astronomy and Astrophysics*, vol. 692, Art. no. A140, EDP, 2024.  
doi:10.1051/0004-6361/202450497.
- “Multiwavelength Investigation of  $\gamma$ -Ray Source MGRO J1908+06 Emission Using Fermi-LAT, VERITAS and HAWC”, Acharyya, A. et al., *The Astrophysical Journal*, vol. 974, no. 1, Art. no. 61, IOP, 2024.  
doi:10.3847/1538-4357/ad698d.
- “A Multiwavelength Study to Decipher the 2017 Flare of the Blazar OJ 287”, Acharyya, A. et al., *The Astrophysical Journal*, vol. 973, no. 2, Art. no. 134, IOP, 2024.  
doi:10.3847/1538-4357/ad64d0.
- “Exploring the Potential of the Pulsed Laser onboard the CALIPSO Satellite to Improve Calibration with VERITAS”, Foote, G. M. et al., in 38th International Cosmic Ray Conference, 2024, Art. no. 1496.  
doi:10.22323/1.444.1496.
- “Indirect search for dark matter with a combined analysis of dwarf spheroidal galaxies from VERITAS”, Acharyya, A. et al., *Physical Review D*, vol. 110, no. 6, Art. no.

063034, APS, 2024.  
doi:10.1103/PhysRevD.110.063034.

- “An Angular Diameter Measurement of  $\beta$  UMa via Stellar Intensity Interferometry with the VERITAS Observatory”, Acharyya, A. et al., The Astrophysical Journal, vol. 966, no. 1, Art. no. 28, IOP, 2024.  
doi:10.3847/1538-4357/ad2b68.
- “VERITAS contributions to the 38th International Cosmic Ray Conference”, Acharyya, A. et al., arXiv e-prints, arXiv:2312.07774, 2023.  
doi:10.48550/arXiv.2312.07774.
- “Multiwavelength Observations of the Blazar PKS 0735+178 in Spatial and Temporal Coincidence with an Astrophysical Neutrino Candidate IceCube-211208A”, Acharyya, A. et al., The Astrophysical Journal, vol. 954, no. 1, Art. no. 70, IOP, 2023.  
doi:10.3847/1538-4357/ace327.
- “Search for Ultraheavy Dark Matter from Observations of Dwarf Spheroidal Galaxies with VERITAS”, Acharyya, A. et al., The Astrophysical Journal, vol. 945, no. 2, Art. no. 101, IOP, 2023.  
doi:10.3847/1538-4357/acbc7b.
- “VTSCat: The VERITAS Catalog of Gamma-Ray Observations”, Acharyya, A. et al., Research Notes of the American Astronomical Society, vol. 7, no. 1, Art. no. 6, IOP, 2023.  
doi:10.3847/2515-5172/acb147.

#### CONFERENCE TALKS & POSTERS

- **“VERITAS Asteroid Occultations: Observations and Considerations”, Poster, ICRC 2025, Geneva, Switzerland, July 2025**
- “VERITAS Optical Programs Overview & Asteroid Occultation Update”, Talk, 2025 Summer VERITAS Collaboration Meeting, University of Utah, June 2025
- “VERITAS Asteroid Occultations: Observations and Considerations”, Poster, 2025 Summer VERITAS Collaboration Meeting, University of Utah, June 2025
- “VERITAS Asteroid Occultations Update”, Talk, 2025 Winter VERITAS Collaboration Meeting, Tucson, Arizona, Jan 2025
- **“ ‘Uncovering truth’ or Occultations with VERITAS: current and future observations”, TNO2024, Taipei, Taiwan, June 2024**
- “Asteroid Occultation Update”, Talk, 2024 Winter VERITAS Collaboration Meeting, Tucson, Arizona, Jan 2024
- “VERITAS Asteroid Occultations Update 2023”, Poster, 2023 Summer VERITAS Collaboration Meeting, UC Santa Cruz, June 2023
- “Resolving sub-milliarcsecond stellar diameters with asteroid occultations at VERITAS”, Poster, Graduate Research Symposium, University of Utah, March 2023
- “VERITAS Asteroid Occultation Overview”, Talk, 2023 Winter VERITAS Collaboration Meeting, Tucson, Arizona, Jan 2023
- **“A blinded search for orbital periodicity in the LS I +61 303 binary system”, Talk, 240th AAS Meeting, Pasadena, California, June 2022**

- “Searching for orbital periodicity in the LS I +61 303 binary system”, Poster, Graduate Research Symposium, University of Utah, March 2022
- “LS I +61 303 orbital period”, Talk, 2022 Winter VERITAS Collaboration Meeting, Virtual, Jan 2022
- ”A blinded search for orbital periodicity in VHE gamma-ray emission in High Mass X-ray Binary LS I +61 303”, Talk, APS 4 Corners Annual Meeting, Oct 2021